STRATA: Resource-Aware Distributed AI Inference on Heterogeneous Computers

ABSTRACT IDEA UPLOAD

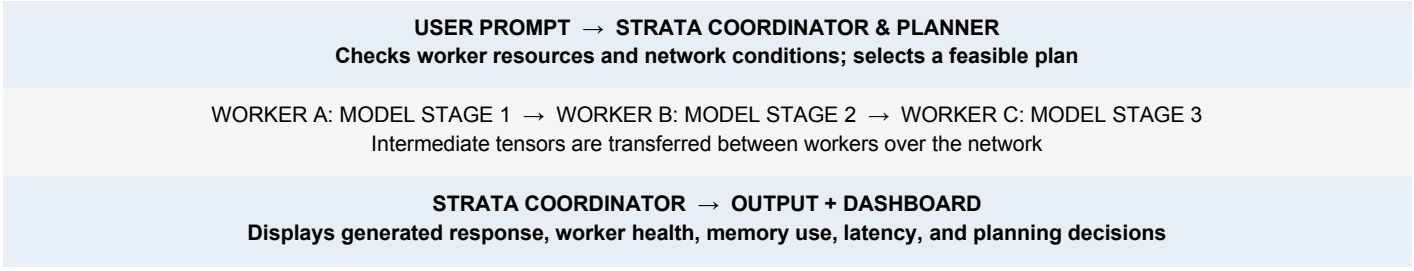
Problem Statement

Running large language models (LLMs) locally can be difficult when a single ordinary computer has limited memory or computing capacity. Using multiple computers introduces additional challenges: each machine has different capabilities, and transferring intermediate model data over a network adds latency and communication overhead. A useful system must consider these trade-offs instead of assuming that distribution will always improve performance.

Methodology

STRATA is a proposed CPU-first distributed inference system. The project will begin with a small supported model and fixed model-stage partitions across two computers, verify correct tensor transfer and output, and then expand to additional workers. A coordinator will collect worker memory and compute availability alongside network measurements, reject plans that exceed resource limits, and compare feasible execution plans. Controlled network delay will be introduced to observe how conditions affect planning. A dashboard will report worker health, memory use, communication latency, inference time, and the reason for each planning decision. Results will be benchmarked against a documented baseline; performance gains will be claimed only if measurements support them.

Proposed Block Diagram



Proposed Outcome

The expected outcome is a demonstrable prototype that runs a supported model across connected computers and makes its planning decisions visible. STRATA aims to combine resource-aware feasibility checks, network-aware plan comparison, and reproducible measurement in a heterogeneous, CPU-based setup. The project will show both when distribution is feasible and when communication costs make it less suitable. This is a proposed project; prototyping has not yet begun, and all outcomes remain to be tested.